$V \otimes W$
$V \oplus W$

$E\ f(g)$ expected value $1/|G| \sum_{g \in G} f(g)$
$g \in G$

==========

=== Representation theory of finite groups ==

https://en.wikipedia.org/wiki/Representation_theory_of_finite_groups

Linear representation (ρ, V) of G
Subrepresentation
Irreducible representation
Irreducible representation: https://en.wikipedia.org/wiki/Irreducible_representation
Let V be a K-vector space.
Let ρ be a map

$\rho: G \to GL(V)$

$\rho(g): V \to V$
$\rho(gg') = \rho(g)\ \rho(g')$ for all $g,g' \in G$

(ρ, V) is a **linear representation** of G.
V is a **representation space** of G
dim(ρ) is the **dimension** of the representation.

A linear representation is simply a case of a group action.
i.e. a **group action** on **vector space V by linear transformation ρ(s)**.

$*: G \times V \to V$ as group action
$g * v = \rho(g)\ v$

$g * = \rho(g): V \to V \in GL(V)$ representation

Two representations (ρ, V) and (ρ', V') may be related by a **linear map of vector spaces V → V'**.
i.e. let $T: V \to V'$ be a linear map such that

$\mathbf{T \circ \rho = \rho' \circ T}$
$T(\rho g\ v) = \rho' g\ (Tv)$ for any $v \in V$

equivalently

$\mathbf{T = \rho' \circ T \circ \rho^{-1}}$

T is called an **equivariant map** or **G-linear map** of ρ and ρ'.

The **identity character** is the simplest 1-dimensional representation

1: G → GL(C)
1(g) = 1 for all g ∈ G

1(g): C → C
1(g) x = x

Clearly, 1 is **irreducible**.

Average values

A recurring theme is to distribute an effect over all G by averaging

Average E [ρ(g)]

The average

$$P: V \to V$$
$$P = \underset{g \in G}{E} [\rho(g)]$$

is a **projection** into the **fixed points of** ρ.
i.e.

projection **P^2 = P**
fixed points **Im P = Fix(ρ)**
$= \{ v \in V: \rho(g)v = v \}$

For any h ∈ G

ρ(h) Pv = ρ(h) Eg [ρ(g)v]
= Eg [ρ(hg)v]
= Pv
=> Im P ⊆ Fix(ρ)

and conversely, for any v ∈ Fix(ρ)

Pv = Eg [ρ(g)v]
= Eg [v]
= v
=> Fix(ρ) ⊆ Im P

Hence Imp P = Fix(ρ). Also

P^2v = Pv since Pv ∈ Fix(ρ)

Invariant E[ρ'(g) ∘ T ∘ ρ(g^-1)]

Let T: V → V' be any linear map.
T is not necessarily an invariant map, but

T': V → V'

$$T' = \underset{g \in G}{E}[\rho'(g) \circ T \circ \rho(g^{-1})]$$

Then T' is always a **invariant map:**

T' ∘ ρ(h) = E[ρ'(g) ∘ T ∘ ρ(g^-1 h)]

$g \in G$

$= E[\rho'(hg') \circ T \circ \rho(g'^{-1})]$ let $g' = h^{-1}g$
$g' \in G$

$= \rho'(h) \circ T'$

When $\rho' = \rho$, then T' can be understood as a **conjugation action**:

$T^{\rho}(g): V \to V$
$T^{\rho}(g) = \rho(g)\, T\, \rho(g^{-1})$

Common representations

Some common representations

character

A **character** i.e. group homomorphism f: G → C
is a **1-dimensional** representation on C

$\rho: G \to GL(C)$
$\rho(g)x = f(g)x$ for any $x \in C$

with **character** equal to itself

$\chi\rho = f$ since $\rho(g)$ has eigenvalue f(g)
=> $tr(\rho(g)) = f(g)$

dual representation (ρ^*, V*) of (ρ, V)

Just **composition** of $f \in V^*$ with $\rho(g)$

$\rho^*: G \to GL(V^*)$

$\rho^*(g): V^* \to V^*$
$\rho^*(g)f = f \circ \rho(g^{-1})$ for any $f: V \to F \in V^*$

Applying on $v \in V$

$\rho^*(g)f: V \to F$
$\rho^*(g)f\,(v) = f\,(\rho(g^{-1})v)$

This preserves the **duality**

$\rho^*(g)f \circ \rho(g) = f \circ \rho(g^{-1}) \circ \rho(g)$
$= f$

direct sum
tensor product

Let $\rho 1: G1 \to GL(V1)$, $\rho 2: G2 \to GL(V2)$ be representations.

direct sum
$\rho 1 \oplus \rho 2$: $G1 \oplus G2 \rightarrow GL(V1 \oplus V2)$
$\rho 1 \oplus \rho 2(g1 \oplus g2) = \rho 1(g1) \oplus \rho 2(g2)$

tensor product
$\rho 1 \otimes \rho 2$: $G1 \otimes G2 \rightarrow GL(V1 \otimes V2)$
$\rho 1 \otimes \rho 2(g1 \otimes g2) = \rho 1(g1) \otimes \rho 2(g2)$

note: for any $f \in GL(V1)$, $f \in GL(V2)$

$f \oplus g(v \oplus w) = f(v) \oplus g(w) \in V \oplus W$
$f \otimes g(v \otimes w) = f(v) \otimes g(w) \in V \otimes W$

Irreducible representations
Let **subspace W** $\subseteq$ **V**
ρ is **G-invariant on W** if

$\rho(g)\ w \in W$ for all $w \in W$

Hence the **subrepresentation** $(\rho|W, W)$ is well-defined.

$\rho|W: W \rightarrow W$

Every representation ρ has two **trivial subrepresentations**:

(ρ, V) itself

$(\rho 0, \{0\})$ **trivial** vector space
i.e. just the 0 vector
$\rho(g)\ 0 = 0$

A representation is **irreducible** if it has no proper subrepresentations
i.e. no non-trivial representations.

The simplest irreducible representation is the identity character **1: G** $\rightarrow$ **GL(C)**

Maschke's Theorem: decomposition
Let (ρ, V) be a representation of G.
If ρ has a subrepresentation in a subspace W of V, then ρ has a subrepresentation in some complement W'.
More precisely

$V = W \oplus W'$
$\rho = \rho|W \oplus \rho|W'$

Hence any representation ρ can be decomposed into a sum of irreducible representations

V = W1 $\oplus$ **...** $\oplus$ **Wn**
ρ **=** ρ**|W1** $\oplus$ **...** $\oplus$ ρ**|Wn** $\rho|Wi$ irreducible

==========

Proof:
Let p: V → V be a projection into W.
i.e. p^2 = p and pw = w

The idea is just to create a projection p' from p.
Let

p': V → V
p' = E[p'^ρ(g)]

Step 1) p' is a **projection into W**:
a) p' preserves W i.e. p'(w) = w for any w ∈ W

p^ρ(g) w = ρ(g) p ρ(g)^-1 w
= w since ρ(g)^-1 w ∈ W and pw' = w'

=> p'(w) = E [p^ρ(g)w]
= E [w]
= w

b) p'^2 = p'
For any v ∈ V

p^ρ(g) v = ρ(g) p ρ(g)^-1 v ∈ W
since p project into W and preserves w

=> p'(v) = 1/|G| ∑ p^ρ(g) v ∈ W

=> p'(p'(v)) = p'(v) since p' preserves W

Thus W' = ker p' is a complement of W.
Step 2) W' is G-invariant
Let v ∈ W'. For any h ∈ G

p'^ρ(h^-1) (v) = p'(v) = 0

=> p'(ρ(h)) v = 0

*) **Decomposition**
Simply by induction on the **dimension** of V.

case: dim V = 1
Trivially irreducible.

case: dim V = n
If ρ is irreducible, then there is nothing to do.
If ρ is invariant on some subspace W ⊆ V, then by the theorem,

V = W ⊕ W'

=> dim W, dim W' < dim V
so
W = W1 ⊕ ... ⊕ Wn ρ|W' irreducible

W' = W1' ⊕ ... ⊕ Wm' ρ|Wi' irredudible

=> V = (W1 ⊕ ... ⊕ Wn) ⊕ (W1' ⊕ ... ⊕ Wm')

QED
==========

Schur's Lemma: uniqueness
https://en.wikipedia.org/wiki/Schur's_lemma
Let (ρ, V) and (ρ', V') be irreducible representation of G.
Then any **non-trivial** map between ρ and ρ' between must be an isomorphism

i.e. for any T: V → V'

T = ρ'^-1 ∘ T ∘ ρ => T is an **isomorphism**
T ≠ 0

Furthermore,

T = T^ρ <=> T is a **scalar** multiple
i.e. **T = λI**

More generally, for any linear map T: V → V', let

T': V → V'
T' = E[ρ'^-1(g) ∘ T ∘ ρ(g)]
g∈G

Then T' is G-linear and hence

case: **ρ',** ρ not isomorphic **T' = 0**

case: **ρ' =** ρ **T' = tr(T) / |G| I**

since E[T^ρ(g)] = λI by above, so taking the trace
=> |G|λ = tr(λI) = E[tr(T^ρ(g))] = E[tr(T)] = tr(T)
=> λ = tr(T) / |G|

In matrix terms:
Let

{aij} i,j = 1...n of ρ
{bst} s,t =1...m of ρ'

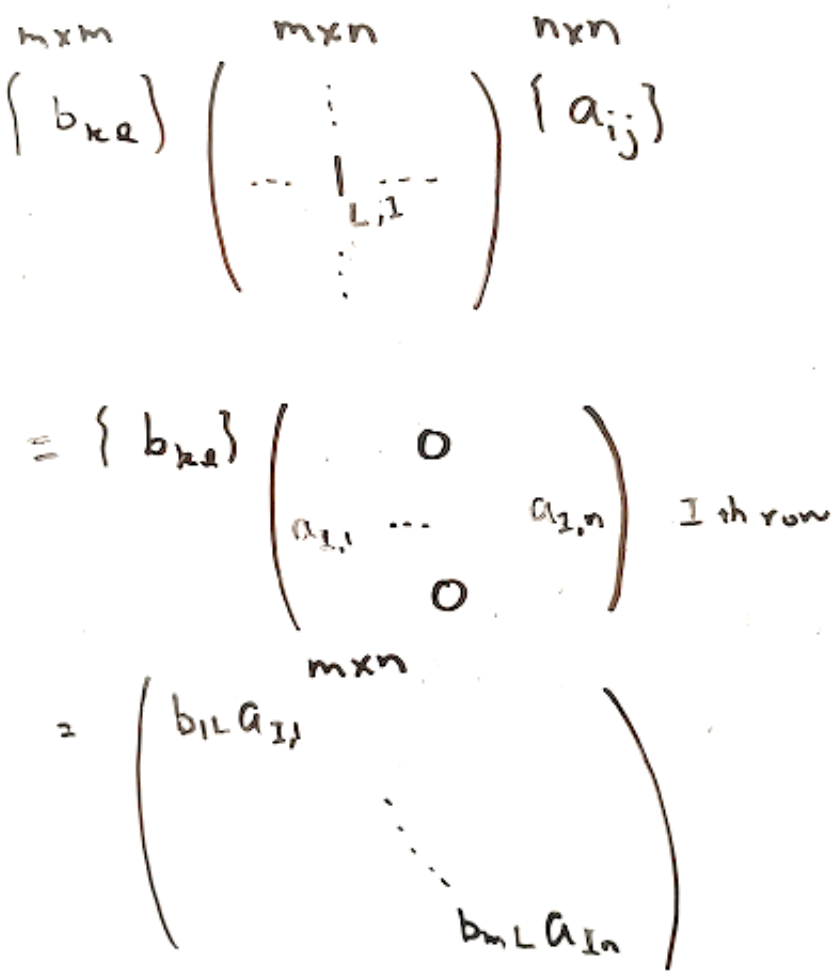
For any i0, t0

1t0,i0 = mxn matrix with 1 in position (t0,i0), 0 otherwise

Then

T' = E[ρ'(g^-1) ∘ 1t0,i0 ∘ ρ(g)]

= { **1/|G|** $\sum$ **bs,t0 (g^-1) ai0,j(g)** } (k,j) th term
for s = 1...m, j = 1...n



Hence, for any i,j, s,t

case: ρ', ρ not isomorphic	bs,t^-1 ● al,j	=	$\sum_{g\in G}$ bs,t (g^-1) ai,j(g)
		=	0
case: $\rho' = \rho$	as,t^-1 ● al,j	=	$\sum_{g\in G}$ s,t (g^-1) al,j (g)
		=	|G|/n if j=k
			0 o.w.

This partitions irreducible representations (ρ, V) of G into **equivalence classes**.
i.e. for any irreducible (ρ, V) of G

equivalence class of (ρ, V) = { all (ρ', V') isomorphic to (ρ, V) }

==========

Proof
Let T: V → V' be a non-trivial map
Then for any v ∈ ker T

$T\rho'(g)\, v = \rho(g)\, Tv$
$= \rho(g)\, 0 = 0$

=> $\rho'(g)\, v \in \ker T$
=> ρ' has a subrepresentation on ker T
=> ker T = {0} or V since ρ' is irreducible
=> ker T = {0} since T is non-trivial
i.e. T is injective

Conversely, for any $w \in \text{Im } T$

$\rho(g)\, w = \rho(g)\, Tv$ for some $w = Tv$
$= T\rho'(g)\, v \in \text{Im } T$

=> ρ has a subrepresentation on Im T
=> Im T = V' or {0} since ρ is irreducible
=> Im T = V since T is non-trivial
i.e. T is surjective

Hence T is bijective.

If V = V', then T has an eigvenvalue $\lambda > 0$.
Let W = eigenspace of λ.
Then for any $w \in W$

$T\rho(g)w = \rho(g)\, Tw$
$= \rho(g)\, \lambda w$
$= \lambda\, \rho(g)\, w$

=> $\rho(g)w$ is also an eigenvector
i.e $\rho(g)w \in W$
=> ρ has a subrepresentation in W
=> W = V

QED
==========

Basic properties
Theorem: Basic properties
Let ρ: G → GL(V) be a representation of a finite group G

Eigenvalues Every $\rho(g\text{^}i)$ has the **same** eigenvector
i.e. if $\rho(g)v = \lambda v$, then

ρ(g^i) v = λ^i v

eigenvalue λ^i
eigenvector v

If g has finite order, then λ is a |g|th **root of unity**

$\lambda\text{^}|g| = 1$ since $v = \rho(1)v = \rho(g\text{^}|g|)v = \lambda\text{^}|g|\, v$

Hence, the sums **cancel** out each other

$$\sum_k \text{tr}(\rho(g^k)) = \sum_k \sum_\lambda \lambda^k$$

$$= \sum_\lambda \sum_k \lambda^k$$

$$= |g| * (\text{multiplicity of } \lambda = 1)$$

by sum of roots of unity
i.e. $\sum \lambda^k = |g|$ if $\lambda = 1$
0 otherwise

==========

=== Character theory ==

https://en.wikipedia.org/wiki/Character_theory

Character
A character is simply the **trace tr(ρ(g))** of a representation.
Let ρ: G → GL(V) be a representation of a finite group.

$\chi\rho$: G → F
$\chi\rho(g) = \text{tr}(\rho(g))$

If the representation ρ is irreducible, then its character $\chi\rho$ is called **irreducible**.

The first important property of the character is that it it **invariant under similarity**
i.e. since

tr(AB) = tr(BA)

Hence

isomorphism invariant

$\chi\rho$ **=** $\chi\rho'$ **if** $\rho \sim \rho'$

since if T is an isomorphism,

$$\chi\rho'(g) = \text{tr}(T \circ \rho(g) \circ T^{-1})$$
$$= \text{tr}(T^{-1} \circ T\, \rho(g))$$
$$= \chi\rho(g)$$

class function

$\chi\rho$ is a class function
i.e.

$\chi\rho(g^x) = \chi\rho(g)$

since $\text{tr}(\rho(g^x)) = \text{tr}(\rho(x)\rho(g)\rho^{-1}(x)) = \text{tr}(\rho(g))$

NOTE: NOT all class functions are representations since class functions are only **semi-multiplicative**

f(gg') = f(gg'^g')
= f(g'g) but not necessarily f(g')f(g)

Due to the additivity of trace, the **decomposition** of representations carries over i.e.

V = V1⊕ ... ⊕Vn
ρ **=** ρ**1** ⊕ **...** ⊕ ρ**n** where ρ: G → GL(V) ρi: G → GL(Vi)
$\chi\rho$ **=** tr(ρ**1** ⊕ **...** ⊕ ρ**n)** where $\chi\rho$: G → C
= $\chi\rho$**1 + ... +** $\chi\rho$**n**

since tr(A⊕B) = tr(A) + tr(B)

In particular, the **irreducible characters** span all characters.
i.e. let **G^** be the set of all irreducible characters

G^ = { $\chi\xi$**1 ....** $\chi\xi$**n }** where ξi is a representative of each **isomorphism class**

Then

$\chi\rho$ **=** $\sum_{\xi \in G^}$ ρ**^(**ξ**)** $\chi\xi$ where p^(ξ) = multiplicity of $\chi\xi$

i.e.

span { $\chi\xi$**:** $\xi \in$ **G^ }** = all characters $\chi\rho$

e.g. identity 1: G → C
Eigenvalue is 1 so

χ**1(g) = 1** for all g ∈ G

The identity is **irreducible**

χ**1** • χ**1** = E[|χ1(g)|^2]
= **1**

e.g. G = n cycle
Let G = (g) and ρ: G → GL(C) be a representation.
Then every representation is a **constant**

ρ(g): C → C
ρ(g)x = cx

and hence

ρ**(g^t) = c^t**

g has order n so

$c^n = \rho(g)^n$
$= 1$

=> c is an **n-th root of unity**

In any case

$\chi\rho(g) = c$
$\chi\rho(g^t) = c^t$

Hence ρ is irreducible

$\chi\rho \bullet \chi\rho = E[\ |\chi\rho(g^j)|^2\ ]$ since |c| = 1 since c is a root of unity
$= E[1]$
$= 1$

<u>Orthonormal basis aka Fourier transform</u>
One of the key property of characters is that the **irreducible characters** form an **orthonormal basis** of not only representations but all **class functions**.
For any f, h: G → C, define the inner product

$\mathbf{f \bullet h = E[f(g)\bar{h}(g)]}$

$= 1/|G| \sum_{g \in G} f(g)\, \bar{h}(g)$

Let L2(G) be the vector space of class functions with the above inner product

f: G → C ∈ L2(G)
f(g^x) = f(g) for any x ∈ G

Then it turns out that the irreducible characters $\chi\xi$ for $\xi \in G^\wedge$ form a **basis** of L2(G)
i.e. each f ∈ L2(G) is a unique linear

$f = \sum_{\xi \in G^\wedge} \xi^*(f)\, \chi\xi$ with coefficients aka **dual**
$\xi^*: L2(G) \to C \in L2(G)^*$

Furthermore, it turns out that the irreducible characters are **orthonormal**.
i.e. for any $\xi, \xi' \in G^\wedge$

$\chi\xi \bullet \chi\xi' = 0$ if $\xi \neq \xi'$ i.e. not isomorphic
$\chi\xi \bullet \chi\xi = 1$

Hence, the duals are precisely the **dot product $\xi^*(f) = f \bullet \chi\xi$**

$f \bullet \chi\xi = (\sum_{\xi' \in G^\wedge} \xi'^*(f)\, \chi\xi') \bullet \chi\xi$
$= \sum_{\xi' \in G^\wedge} \xi'^*(f)\, (\chi\xi' \bullet \chi\xi)$
$= \xi^*(f)$ by orthonormality

aka **Fourier inversion**

For any class functions f, f' $\in$ L2(G)

$$f \bullet f' = \underset{g \in G}{E}[f(g)\bar{f'}(g)]$$

$$f \bullet f' = \sum_{\xi \in \hat{G}} \xi^*(f)\, \chi_\xi \bullet f' = \sum_{\xi \in \hat{G}} \xi^*(f)\xi^*(f')$$

Hence, the **duality**

$$\mathbf{\underset{g \in G}{E}[f(g)\bar{f'}(g)] = f \bullet f' = \sum_{\xi \in \hat{G}} \xi^*(f)\xi^*(f')}$$

In particular, for f' = f, the **dual norm** aka Parseval identity

$$\mathbf{\underset{g \in G}{E}[|f(g)|^2] = f \bullet f = \sum_{\xi \in \hat{G}} \xi^*(f)^2}$$

Often written as

$$\underset{\text{L2 norm}}{\| f \|} = \underset{\text{l2 norm}}{\| f^* \|}$$

In particular, if f is a character i.e. f = χ_ρ, then the Fourier coefficients are exactly the **multiplicity** of the irreducibles
i.e. if

$\rho = \bigoplus_i \rho_i$ ρ_i irreducible

and

k_ξ = # of $\rho_i \sim \xi$

Then

$$\chi_\rho = \sum_i \chi_{\rho_i} = \sum_{\xi \text{ over all } \rho_i \sim \xi} (\sum 1)\, \chi_\xi = \sum_{\xi \in \hat{G}} k_\xi\, \chi_\xi$$

Hence

$\xi^*(\rho) = k_\xi \in \mathbf{N}$

and

$$\chi_\rho \bullet \chi_\rho = \sum \xi^*(\rho)^2 = \sum_{\xi \in \hat{G}} \mathbf{k_\xi^2}$$

In fact, this is an exact condition for when a class function is a character
Class function f: G → C is a **character** for some representation if and only if the coefficients $\xi^*(f)$ are all **integers**.
i.e.

f **is a character** <=> $\xi^*(f) \in N$ for all ξ since =>) if $\rho = \oplus$ ni ξi, then $\chi\rho = \sum$ ni $\chi\xi$i

i.e. f = $\chi\rho$ ni = $\chi\rho \bullet \chi$i = $\xi^*(\rho)$

<=) let $\rho = \oplus \xi^*(f)\ \xi$
Then $\chi\rho = \sum \xi^*(f)\ \chi\xi$

<=> f ● f' ∈ N for any character f' since if f' is a character, then f ● f' = $\sum \xi^*(f')\ \xi^*(f'') \in N$

And irreducible characters:

ρ **is irreducible** <=> $\chi\rho \bullet \chi\rho = 1$ since =>) by orthogonality
<=) $1 = \chi\rho \bullet \chi\rho = \sum |\xi^*(\rho)|^2$ where each $\xi^*(\rho)$ is an integer

Hence, $\xi 0^*(\rho) =$ 1 for exactly one $\xi 0$ and

$\rho = \sum \xi 0^*(\rho)\ \xi = \xi 0$

<u>Identity 1G: G → C</u>
The identity representation 1: G → C is a special type of **irreducible** representation.

$\sum_{g \in G} |\chi\xi(g)|^2$ = |G| E[$|\chi\xi(g)|^2$]
= |G| $\chi\xi \bullet \chi\xi$
= |G| by orthogonality

whereas

$\sum_{g \in G} \chi\xi(g)$ = |G| E[$\chi\xi(g)$]
= tr($\sum \xi(g)$)

= |G| if $\chi = 1$ See basic properties below for proof
0 o.w.

i.e. the characters $\chi\xi$ cancel out in the average of $\chi\xi(g)$ but not in the norm $|\chi\xi(g)|$

Hence

representation	$\sum$ **1(g)**	**= |G| 1(g)**
character	$\sum \chi$**1(g)**	**= |G|**

Hence, for any class function f ∈ L2(G), the average is determined **completely by 1**.

$$\sum_{g \in G} f(\mathbf{g}) = \sum_{g \in G} \sum_{\xi \in G\hat{}} \xi^*(f)\, \chi\xi(g)$$

$$= \sum_{\xi} \xi^*(f) \sum_{g} \chi\xi(g)$$

$$= 1^*(f) \qquad \text{all 0 except for } \xi = 1$$

$$= f \bullet \mathbf{1}$$

==========

Proofs

*) Orthogonality

$$\chi\rho \bullet \chi\rho' = E[\chi\rho(g)\chi\rho'^{-}(g)]$$

$$= E[\chi\rho \otimes \rho'\hat{}^{-1}(g)]$$

$$= \dim \mathrm{Fix}(\rho \otimes \rho'\hat{}^{-1})$$

If $\rho \neq \rho'$, then Fix $(\rho \otimes \rho'\hat{}^{-1}) = \varnothing$ since ρ and ρ'

TODO

If $\rho = \rho'$ then

$\rho(g) \otimes \rho(g)\hat{}^{-1}\ v \otimes w = \sum$

*) **Basis**

Let f be a class function and

$$p = f - \sum \xi^*(f)\, \chi\xi$$

Then

$$p \bullet \chi\xi = f \bullet \chi\xi - \sum \xi'^*(f)\, \chi\xi' \bullet \chi\xi$$
$$= 0$$

If the irreducibles do not form a basis, then there exists an f such that $p \neq 0$ but $p \bullet \chi\xi = 0$ for all ξ.

Furthermore, since $G\hat{}$ spans the characters $\chi\rho$,

$$p \bullet \chi\rho = p \bullet \sum \xi^*(\rho)\, \chi\xi$$
$$= \sum \xi^*(\rho)\, p \bullet \chi\xi$$
$$= 0$$

Now, **apply** this to the representations. Let

$$p\rho: V \to V$$
$$p\rho = \sum_{g \in G} p(g)\, \rho(g)$$

There is an adjoint relation

$p\rho\ \rho(x)$ $= \sum_{g\in G} p(g)\ \rho(gx)$

$= \sum_{g\in G} p(g)\ \rho(x)\ \rho(g\hat{}x)$

$= \sum_{g'\in G} p(g'\hat{}x)\ \rho(x)\ \rho(g')$ sum over g' = g^-x

$= \rho(x)\ p\rho$ p(g^x) = p(g) since p is a class function

=> $p\rho$ is some scalar λI

=> $|G|\lambda = tr(p\rho) = p \bullet \chi\rho^{-} = 0$

=> $p\rho = 0$

i.e. $p\rho$ is **always** the zero map, which certainly cannot be.

Indeed, with ρ = R, the regular map

$0 = pR[1]$

$= \sum_{g\in G} p(g)\ [g] \neq 0$ since p(g) ≠ 0 for at least one g

QED

*) **average** $\sum \chi\xi$

Recall that

$P: G \to GL(V)$

$P = \sum \rho(g)$

is a projection into $Fix(\rho) = \{ v \in V: \rho(g)v = v \}$.

Hence the eigenvectors are precisely

$Pv = v \quad v \in Fix(\rho)$

=> $tr(P) = \sum 1$

$= \dim Fix(\rho)$

But

$\rho(g)v = v \in Fix(\rho)$

=> $\rho|Fix(\rho)$ is a subrepresentation

=> $Fix(\rho) = \varnothing$ if ρ is irreducible, except ρ=1

Hence, if $\xi \neq 1$

$\sum \chi\xi = tr(P)$

$= \dim Fix(\xi)$

$= 0$

QED

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Theorem: Basic properties

Basic

Identity $\chi\rho(1) = \dim V$

inverse $\chi\rho(g^{-1}) = \chi\rho(g)^{-1}$
$= \chi\rho(g)^{-}$ since the eigenvalues are roots of unity
$\mathrm{tr}(\rho(g)^{-1}) = \sum 1/\lambda = \sum \lambda^{-} = \mathrm{tr}(\rho(g))^{-}$

sum $\chi\rho + \chi\rho'$ $= \mathrm{tr}(\rho 1)+\mathrm{tr}(\rho 2)$
$= \mathrm{tr}(\rho 1 \oplus \rho 2)$ since $\mathrm{tr}(A \oplus B) = \sum aii + \sum bjj + \mathrm{tr}(A) + \mathrm{tr}(B)$
$= \chi(\rho 1 \oplus \rho 2)$

product $\chi\rho \cdot \chi\rho'$ $= \mathrm{tr}(\rho)\,\mathrm{tr}(\rho')$
$= \mathrm{tr}(\rho \otimes \rho')$ since $\mathrm{tr}(A \otimes B) = \sum i,j\ aii\ bjj = \mathrm{tr}(A)\,\mathrm{tr}(B)$
$= \chi(\rho \otimes \rho')$

Average

$$\mathbf{E}[\chi\rho(g)] = \dim \mathrm{Fix}(\rho)$$
$$g \in G$$

where $\mathrm{Fix}(\rho) = \{ v: \rho(g)v = v \text{ for all } G \}$
Note also that

$(\rho, \mathrm{Fix}(\rho))$

is a subrepresentation since $\rho(g)v = v \in \mathrm{Fix}(\rho)$ for any $v \in \mathrm{Fix}(\rho)$

Hence, if ρ is **irreducible**, then either

$\mathbf{E}[\chi 1(g)] = 1$
$\mathbf{E}[\chi\rho(g)] = 0$ **if** $\rho \neq 1$ since $\mathrm{Fix}(\rho)$ must be $\varnothing$ by irreducibility

invariance

Invariant under **conjugation**
i.e. characters are class functions

$\chi\rho(g) = \chi\rho(g^x)$

Invariant under **isomorphism**
i.e. if $\rho \sim \rho'$, then

$\chi\rho = \chi\rho'$

Irreducible characters
Let ξ be irreducible representations

orthonormal $\chi\xi \bullet \chi\xi' = 0$ if $\xi \neq \xi'$
$\chi\xi \bullet \chi\xi = 1$

average $\sum_{g\in G} \chi\xi(g) = 0$ if $\xi \neq 1$
$\sum_{g\in G} \chi 1(g) = |G|$

L2 norm $\sum_{g\in G} |\chi\xi(g)|^2 = |G|\ \chi\xi \bullet \chi\xi = |G|$

Dual sums over G^

dual orthonormal classes i.e. $g' \notin g^G$ $\sum \chi\xi(g)\chi\xi^{-}(g') = 0$ if $g \not\sim g'$ by conjugacy

$\sum_{\xi\in G^} \chi\xi(g)\chi\xi^{-}(g') = \sum |\chi\xi(g)| = |C(g)|$ if $g \sim g'$

note: $C(g)$ = center of $g = \{ x: gx = xg \}$

note: this is just the function $\varphi(x) = 1$ for a single conjugacy class $x\in g^G$ and 0 o.w.

$$\varphi = \sum \xi^*(\varphi)\ \chi\xi = 1/|C(g)| \sum \xi^{-}(g)\ \chi\xi$$

Orthonormal Basis
Irreducible characters form a basis of class functions

Basis

$$f = \sum \xi^*(f)\ \chi\xi = \sum (f \bullet \chi\xi)\ \chi\xi$$

where

$\xi^*: G \to C \in G^*$
$\xi^*(f) = f \bullet \chi\xi$

Thus the dimension of class functions L2(G) is

dim (L2(G)) = |G^|
= # of conjugacy classes

Parseval
For any class functions $f, f' \in L2(G)$

$$E[\ f(g)f'(g)] = f \bullet f' = \sum_{\xi\in G^} \xi^*(f)\ \xi(f)$$
$g\in G$

For f = f'

$$E[\ |f(g)|^2] = f \bullet f = \sum_{\xi\in G^} \xi^*(f)^2$$
$g\in G$

Characterization
If

$\rho = \oplus c\xi\ \xi \qquad c\xi \in N$

then

$\chi\rho \bullet \chi\rho \qquad = \sum c\xi\text{^}2$

where

$\xi^*(f) \qquad = c\xi \in N$

i.e. all coefficients $\xi^*(f)$ are integers

Hence the characterization

f is a character <=> $\xi^*(f) \in N$ for all ξ

<=> f ● f' ∈ N for any character f'

ρ **is irreducible** <=> $\chi\rho \bullet \chi\rho = 1$
i.e. $\rho \in G\text{^}$

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Proofs

*) Orthogonality

$\chi\rho \bullet \chi\rho' = E[\chi\rho(g)\chi\rho'^{-}(g)]$

$= E[\chi\rho\otimes\rho'\text{^}-1\ (g)]$

$= \dim Fix(\rho\otimes\rho'\text{^}-1)$

If $\rho \neq \rho'$, then Fix $(\rho\otimes\rho'\text{^}-1) = \varnothing$ since ρ and ρ'

TODO

If $\rho = \rho'$ then

$\rho(g)\otimes\rho(g)\text{^}-1\ v\otimes w = \sum$

*) Basis

*) **average** $\sum \chi\rho(g)$
Let

$P: G \rightarrow GL(V)$
$P = \sum \rho(g)$

Recall that P is a projection into $Fix(\rho) = \{ v \in V: \rho(g)v = v \}$.
Then

$\sum \chi\rho(g)$ = tr($\sum\rho(g)$)
= tr(P)

Hence, it suffices to compute the trace of P.
The eigenvectors are precisely

Pv = v $v \in Fix(\rho)$

=> tr(P) = $\sum 1$
= dim $Fix(\rho)$

QED

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=== Induced characters ==

Induced class function f^G: G → F
Restriction f|H: H → F
The induced class function is simply the simplest way to **extend aka induce** a class function from a **subgroup** H to a larger group G
Let $f \in L2(H)$. Then **extend (induce)** an class function **f^G** $\in$ **L2(G)** by

f^G: G → C
f^G(g) = 1/|H| $\sum_{x \in G}$ f'(g^x) where f'(x) = f(x) if $x \in H$
0 o.w.

i.e. just combine all possible **conjugations g^x**
f^G $\in$ **L2(G)** is said to be the function **induced by f** $\in$ **L2(H)**.

Indeed, f^G is a class function since the conjugation is absorbed

f^G(g^y) = 1/|H| $\sum_{x \in G}$ f'(g^yx) g^y absorbed by sum
= **f^G(g)**

This definition ensures that if G = H, then **f^H = f**
i.e.

f^H(h) = 1/|H| $\sum$ f(h^x)
= 1/|H| $\sum_{h \in H}$ f(h)
= f(h)

Conversely, if $f \in L2(G)$ is a class function, then the **restriction f|H** $\in$ **L2(H)** is just a restriction to H

f|H: H → F
f|H = f

Duality
There is a dual relationship
Let $f \in L2(H)$ and $\varphi \in L2(G)$ be class functions. Then

Frobenius reciprocity

f^G ●G φ **= f ●H** φ**|H** where ●G = 1/|G| ∑ g ∈ G __
●H = 1/|H| ∑ h ∈ H __

since

f^G ●G φ = 1/|G||H| ∑ f^(g^x) φ^{-}(g)
g,x∈G

= 1/|G||H| ∑ f^(h) φ^{-}(h^-x) with h = g^x
h∈H,x∈G

= 1/|H| ∑ f^(h) φ^{-}(h) since φ(h^x) = φ(h) by invariance
h∈H
= **f ●H** φ**|H**

This duality is called **Frobenius reciprocity**.

Characters are preserved by the induction.
i.e.

f ∈ L2(H) is a **character** => f^G ∈ L2(G) is a **character**

since for any irreducible ξ∈G^

ξ*(f^G) = f^G ●G $\chi\xi$
= f ●H $\chi\xi$|H ∈ N since both f and $\chi\xi$|H are both characters in H

This implies that f^G is **determined** by the coefficients f ●H $\chi\xi$|H
i.e. if f ∈ L2(H) is a class function, and

φ ∈ L2(G) is a class function => φ **= f^G** since ξ*(φ) = f ● $\chi\xi$|H = ξ*(f^G)
ξ*(φ) = φ|H ●H $\chi\xi$|H

Theorem: Basic properties

Basic
If f ∈ L2(G) and H ◁ K ◁ G, then

f^G = (f^K)^G

Reciprocity f^G ●G φ = f ●H φ|H

Character f ∈ L2(H) is a character => f^G ∈ L2(G) is a character

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=== Regular representation ===

Regular representation
https://en.wikipedia.org/wiki/Regular_representation
The regular representation is effectively just a **permutation** of G itself.
The vector space is the free vector space F[G] of G itself
i.e. elements are **free sum i.e. formal sum** of $g \in G$

$v \in F[G]$

$\mathbf{v = \sum_{x \in G} kx\,[x]}$ for coefficients $kx \in F$

$\dim F[G] = |G|$

The **regular representation R** of G is

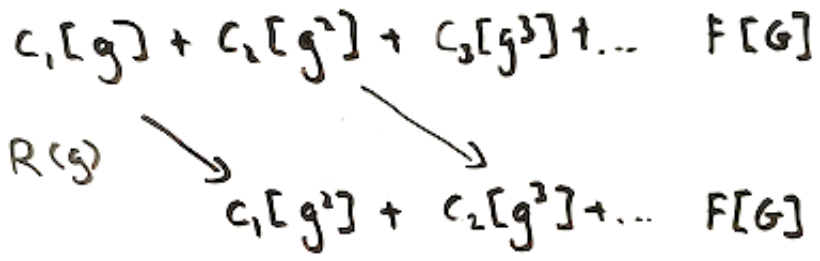
R: G → GL(F[G])
R(g): F[G] → F[G]
R(g) = [g *]

Applied on a basis $[x] \in F[G]$

R(g): F[G] → F[G]
R(g)([x]) = [gx]

Hence

$R(g)(\sum kx\,[x]) = \sum kx\,[gx]$



i.e. a **permutation**
Since R has a very large degree |G|, it can be decomposed.

<u>Subrepresentation G|H</u>
Let $H \lhd G$
Then the **cosets** $xH \in G|H$ form a **subrepresentation of R**
i.e. Let

element **[xH]** $= \sum_{\mathbf{h} \in \mathbf{H}} \mathbf{[xh]} \in F[G]$ for any $x \in G$

subspace **F[G|H]** = subspace generated by [xH] for all $xH \in G|H$

dim F[G|H]= [G:H] = |G|/|H|

Then R|(G|H) is a subrepresentation.
For any $g \in G$

R(g)([xH]) $= \sum_{h \in H} [gxh]$

= [gxH] $\in$ **F[G|H]**

For each x $\in$ G, [xH] is precisely a fixed point for g $\in$ H^x

gxH = xH <=> g $\in$ H^x

Hence

eigenvalues λ = 1
R(g)[xH] = [xH] if g $\in$ xH

character r(g) = | { xH: g $\in$ H^x }|
= 1/|H| | { x $\in$ G: g^x $\in$ H } | 1/|H| for multiplicity
i.e. if g^x $\in$ H, then g^(xH) $\in$ H

r(1) = dim F[G|H]
= [G:H]

Some special cases are:

Normal H
i.e. H = N(H)
Then

r(h) = dim H' h $\in$ H
= [G:H]

r(g) = 0

hence

r ● r = E[|r(g)^2]
= 1/|G| (∑ |r(h)^2|)
= 1/|G| |H| [G:H]^2

= [G:H]

H is reducible since r ● r > 1

Malnormal
i.e. H^x ∩ H^y = {1} for any x ∉ yH
Then

r(g) = 1 for g $\in$ G/K, g ≠ 1
r(k) = 0 for k $\in$ K

Hence

r ● r = E[|r(g)|^2]
g$\in$G
= 1/|G| (|r(1)|^2 + ∑ |r(k)^2| + ∑ |r(g)^2|)
k $\in$ K g $\in$ G/K
= 1/|G| ([G:H]^2 + 0 + ∑ 1

= 1/|G| ([G:H]^2 + |G| - [G:H])

= 1 + ([G:H] - 1) / |H|

H is reducible since r ● r > 1

Consider the trace of R(g): G → GL(C[G])
R(g) is just a **permutation by g** so has no fixed points except g = 1

$\chi R(g)$ = tr(R(g))
= 0 sum of diagonals = 0
$\chi R(1)$ = |G|

Hence the coefficients are exactly

$\chi R \bullet \chi\xi = 1/|G| \sum \chi R(g)\, \chi\xi(g)$

$= \chi\xi(1)$
= **dim V**ξ > 0 where representation (ξ, Vξ)

The regular representation thus contains **all irreducible characters**

χ**R** $= \sum \chi\xi(1)\, \chi\xi$

$= \sum_{\xi \in G\hat{}} \textbf{dim V}\xi \;\; \chi\xi$

as expected, since R is the freest representation.

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=== Examples ===

Example: cyclic
e.g. G = Z4 represented in C* = unit cycle
Let

ρ: G → GL(C*)

Then

ρ**(n)** ∈ **{ 1, -1, i, -i }** since every n ∈ Z4 has degree 4
i.e. ρ(n)^4 = 1

ρ**(1)** determines ρ since 1 is a generator of Z4

Hence, possible representations are

$\rho(0) = 1$ $\rho(1) = i$ $\rho(2) = -1$ $\rho(3) = -i$

$\rho(0) = 1$ $\rho(1) = -i$ $\rho(2) = -1$ $\rho(3) = i$

$\rho(0) = 1$ $\rho(1) = -1$ $\rho(2) = 1$ $\rho(3) = -1$

Example: Dihedral group
https://groupprops.subwiki.org/wiki/Linear_representation_theory_of_dihedral_groups

Recall that the Dn is defined by n-1 rotations and a reflection

α**^n**	= 1	rotation identity
β**^2**	= 1	reflection identity
$\alpha\beta\alpha\beta$	= 1	

|Dn| = **2n** since α^k n elements
$\alpha^k\beta$ n elements

conjugacy classes = **(n+3) / 2** if n odd
(n+6) / 2 if n even

Let $\theta = 2\pi/n$
A representation ρ: Dn → GL(R^2) is:

ρ: Dn → GL(R^2)

	matrix	**eigenvalues**	**character**
$\rho(\alpha^k)$	= |cos kθ -sinkθ | | sin kθ coskθ|	e^ikθ, e^-ikθ	$\chi\rho(\alpha^k) = 2\cos k\theta$
$\rho(\beta)$	= |0 -1| |1 0 |	0	$\chi\rho(\beta) = 0$

hence also

$\rho(\alpha^k\beta)$	= $\rho(\alpha^k)\rho(\beta)$ = |-sin kθ cos kθ | | cos kθ sin kθ|	0	$\chi\rho(\alpha^k\beta) = 0$

ρ is clearly **irreducible**, since if ρ|W is a non-trivial subrepresentation, then

1 < dim W < dim R^2 = 2

which is impossible.

Hence, the norm of the characters should also be 1

1 = $\chi\rho \bullet \chi\rho$

= $\underset{x\in Dn}{E}[\ |\chi\rho(x)|^2\ ]$

= $1/|Dn| \sum_{k=0...n-1} 4\cos^2 k\theta$ $2\cos^2\theta = 1 + \cos 2\theta$

= $1/|Dn| \sum_{k=0...n-1} 2\ (1 + \cos 2k\theta)$

= $1 + 2/|Dn| \sum_{k=0...n-1} \cos 2k\theta$

= 2 if n = 2
1 if n > 2

Concretely

	|Dn|	θ	$\chi\rho \bullet \chi\rho = 1$
D2	4	π	1/4 (χ1^2 + \|$\chi\rho(\alpha)$|^2) = (1 + 1) = 2 Special case where D2 is **reducible**
D3	6	$2\pi/3$	1/6 (χ1^2 + \|$\chi\rho(\alpha)$\|^2 + \|$\chi\rho(\alpha$^2)|^2) = 4/6 (1 + (1/2)^2 + (1/2)^2) since cos $2\pi/3$ = 1/2 = 1
D4	8	$\pi/2$	1/8 (χ1^2 + \|$\chi\rho(\alpha)$\|^2 + \|$\chi\rho(\alpha$^2)\|^2 + \|$\chi\rho(\alpha$^3)|^2) = 1/2 (1 + 0 + 1 + 0) = 1

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== OLD ==

Characters are invariant under **isomorphism**.
i.e. if ρ' is isomorphic to ρ, then

$\chi\rho$'(g) = tr(T ∘ ρ(g) ∘ T^-1) where ρ' ∘ T = T ∘ ρ
= tr(T^-1 ∘ T ρ(g))
= $\chi\rho$(g)

Characters are **class functions** since traces are similarity invariant.
note: a class function is **invariant under conjugation** f(g^x) = f(g) for any x ∈ G
i.e. for any x ∈ G

$\chi\rho$(g^x) = tr(ρ(g^x))
= tr(ρ(x) ρ(g) ρ^-1(x))
= $\chi\rho$(g) by similarity invariance tr(B^-1AB) = tr(A)

Not all class functions are representations since class functions are only **semi-multiplicative**

f(gg') = f(gg'^g')
= f(g'g) but not necessarily f(g')f(g)

Due to the trace being **additive** tr(A+B) = tr(A) + tr(B) and **semi-multiplicative** tr(AB) = tr(BA)
properties of representations carry over to the character:

arithmetic

additive $\chi\rho+\chi\rho'$ $= \chi(\rho\oplus\rho')$

semi multiplicative $\chi\rho\cdot\chi\rho'$ $= \chi(\rho\otimes\rho')$

orthonormal

if (ξ, V), (ξ', V') are irreducible representations of G, then

$\chi\xi \bullet \chi\xi' = 1$ if $\xi = \xi'$
0 o.w.

decomposition

if $\rho = \rho 1 \oplus ... \oplus \rho n$ where $(\rho i, Vi)$ are representations
then

$\chi\rho = \chi\rho 1 + ... + \chi\rho n$ since $\chi\rho+\chi\rho' = \chi(\rho\oplus\rho')$

Hence if ξi is a set of all irreducible representations, then every representation has the form

$$\chi\rho = \sum i\ (\chi\rho \bullet \chi\xi i)\ \chi\xi i$$

$\chi R(g) = 0$
$\chi R(1) = |G|$
$E[\chi R(g)] = 1$

==========

proof:

- Average $E[\chi\rho(g)]$

Let P be the average map

P: V → V
P = 1/|G| ∑ ρ(g)

Then

$E[\chi\rho(g)] = 1/|G| \sum tr(\rho(g))$
$= 1/|G|\ tr(P)$

claim: P is a projection into $Fix(\rho)$.
i.e. P^2 = P

If the claim is true, then

tr P = dim (Im P)
= dim $Fix(\rho)$

Proof claim.
For any $v \in Fix(\rho)$

$$\begin{aligned} Pv &= 1/|G| \sum \rho(g)\, v \\ &= 1/|G| \sum v \\ &= v \end{aligned}$$

$$\Rightarrow Fix(\rho) \subseteq Im\ P$$

Conversely, for any v

$$\begin{aligned} \rho(g')\, Pv &= 1/|G| \sum \rho(g'g)\, v \\ &= Pv \end{aligned}$$

$$\Rightarrow Im\ P \subseteq Fix(\rho)$$

Hence $Im\ P = Fix(\rho)$.
Furthermore, by above

$Pv = v$ for any $v \in Fix(\rho)$
$\Rightarrow PPv = Pv$ since $Pv \in Im\ P = Fix(\rho)$

Hence **P^2 = P** i.e. a projection
QED claim

QED
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This
is
code
Monaco 14